

Chapter 07: 802.11n (Wi-Fi 4) - MIMO Revolution

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CS6206 Selected Topics in Wireless Networks



Lecture 8 Learning Objectives

By the end of this lecture, you will be able to:

- Explain MIMO concepts: spatial streams, spatial multiplexing, beamforming
- Calculate throughput gains from channel bonding (20→40 MHz)
- Describe frame aggregation mechanisms (A-MSDU and A-MPDU)
- Explain block acknowledgments and their efficiency benefits
- Analyze HT (High Throughput) PHY and MAC enhancements
- Calculate maximum theoretical throughput for various 802.11n configurations
- Understand backward compatibility mechanisms with 802.11a/b/g

802.11n: The Quantum Leap (2009)

- **Ratification:** 2009 (draft products from 2007)
- **Wi-Fi Generation:** Wi-Fi 4 (retroactively named)
- **Key Innovation:** MIMO (Multiple Input Multiple Output)
- **Maximum PHY Rate:** 600 Mbps (theoretical)
- **Practical Throughput:** 100-300 Mbps
- **Frequency Bands:** 2.4 GHz and 5 GHz (first dual-band standard)

Four Pillars of 802.11n:

- 1 **MIMO:** Multiple antennas for spatial multiplexing
- 2 **Channel Bonding:** 40 MHz channels (2×20 MHz)
- 3 **Frame Aggregation:** Combine multiple frames
- 4 **Block Acknowledgments:** Acknowledge multiple frames at once

MIMO Fundamentals: Beyond Single Antenna

- **SISO (Single Input Single Output):** Traditional 1×1
- **MIMO (Multiple Input Multiple Output):** $N \times N$ antennas
- **Spatial Streams:** Independent data flows ($\max = \min(\text{Tx antennas, Rx antennas})$)
MIMO means Multiple Input Multiple Output: multiple transmit and receive antennas are used together.

Two Key MIMO Benefits: Spatial diversity means sending redundant copies of the signal so the receiver can choose the best one, improving reliability and robustness.

- ① **Spatial Diversity:** Multiple copies, pick best (improves reliability)
- ② **Spatial Multiplexing:** Multiple streams simultaneously (increases throughput)
Spatial multiplexing means sending different data streams simultaneously over different antennas, increasing throughput

Example Configurations:

- **2×2 MIMO:** 2 transmit, 2 receive $\rightarrow$ up to 2 spatial streams
- **3×3 MIMO:** 3 transmit, 3 receive $\rightarrow$ up to 3 spatial streams
- **4×4 MIMO:** 4 transmit, 4 receive $\rightarrow$ up to 4 spatial streams

Limiting Factor: Client devices often have fewer antennas than APs

Numerical Example 1: MIMO Throughput Gain I

Problem: Compare throughput for:

- 802.11g (SISO): 54 Mbps PHY, 20 Mbps practical TCP
- 802.11n 2×2 MIMO: 2 spatial streams, each 65 Mbps, 40 MHz channel

Calculate:

- ➊ Maximum PHY rate for 802.11n configuration
- ➋ Practical TCP throughput improvement factor
- ➌ Effect of client having only 1 antenna

Given:

- 802.11n rate scaling: Each spatial stream ~ 65 Mbps in 40 MHz with 64-QAM
- Protocol efficiency: $\sim 50\%$ for 802.11g, $\sim 60\%$ for 802.11n (due to aggregation)

Numerical Example 1: MIMO Throughput Gain II

- 2×2 MIMO: 2 spatial streams possible if both ends have 2 antennas

Solution Steps:

- ① 802.11n PHY rate: $2 \text{ streams} \times 65 \text{ Mbps/stream} = 130 \text{ Mbps}$
- ② 802.11g practical: $54 \text{ Mbps} \times 50\% = 27 \text{ Mbps TCP}$
- ③ 802.11n practical: $130 \text{ Mbps} \times 60\% = 78 \text{ Mbps TCP}$
- ④ Improvement: $78/27 = 2.9\times$ faster
- ⑤ With 1-antenna client: Only 1 stream possible = 65 Mbps PHY, 39 Mbps TCP
- ⑥ Still improvement: $39/27 = 1.44\times$ faster than 802.11g

Model Answer:

- ① Maximum PHY rate: 130 Mbps ($2 \times 65 \text{ Mbps}$)
- ② TCP throughput improvement: $2.9\times$ faster (78 vs 27 Mbps)
- ③ With 1-antenna client: $1.44\times$ faster (39 vs 27 Mbps) - still improvement from channel bonding and aggregation

Channel Bonding: 20→40→(optional 40+40) MHz I

- **802.11a/g:** 20 MHz channels
- **802.11n:** Optional 40 MHz channels (bond two 20 MHz channels)

Benefits:

- **Theoretical:** $2\times$ throughput (not exactly $2\times$ due to guard bands)
- **Actual:** $\sim 1.8\text{-}1.9\times$ throughput gain

Drawbacks:

- **Fewer channels:** 2.4 GHz: $3\times 20\text{ MHz} \rightarrow 1\times 40\text{ MHz} (+1\times 20\text{ MHz leftover})$
- **More interference:** Wider channel catches more interference
- **Regulatory:** Some regions restrict 40 MHz in 2.4 GHz

40 MHz Channel Types:

- **Contiguous:** Two adjacent 20 MHz channels
- **Non-contiguous (40+40):** Two separate 20 MHz channels (rarely used)

Channel Bonding in 2.4 GHz Band

802.11a/g used 20 MHz channels. 802.11n can optionally bond two adjacent 20 MHz channels into a 40 MHz channel. In theory that should double throughput, but in practice the gain is more like 1.8–1.9× because of guard bands and overhead.

2.4 GHz Reality:

- Only 3 non-overlapping 20 MHz channels (1, 6, 11)
- Bonding channels 1+5 or 6+10 creates 40 MHz channel
- But: Channels 5 and 10 overlap with other channels!
- **Result:** 40 MHz in 2.4 GHz causes massive interference

Recommendation: Avoid 40 MHz in 2.4 GHz, use only in 5 GHz

Frame Aggregation: Reducing MAC Overhead

- **Problem:** MAC/PHY overhead dominates for small frames
- **802.11a/g overhead:** $\sim 50\text{-}100\ \mu\text{s}$ per frame regardless of size
- **Solution:** Send multiple frames in one transmission

Two Aggregation Methods:

① A-MSDU (Aggregate MAC Service Data Unit):

- Combine multiple MSDUs into one MPDU
- Single MAC header, multiple payloads
- Max size: 7935 bytes (up from 2304)

② A-MPDU (Aggregate MAC Protocol Data Unit):

- Combine multiple MPDUs into one PHY transmission
- Each has its own MAC header
- Max size: 65535 bytes

Modern implementations use both: A-MPDU containing A-MSDUs

A-MSDU vs A-MPDU Comparison

Aspect	A-MSDU	A-MPDU
What's aggregated	Multiple MSDUs (Layer 2 frames)	Multiple MPDUs (MAC frames)
Headers	Single MAC header for all	Each MPDU has own MAC header
Max Size	7935 bytes	65535 bytes
Error Recovery	All or nothing - retransmit entire A-MSDU	Individual MPDUs can be retransmitted
Efficiency	Higher (less header overhead)	More robust (selective retransmission)
Common Use	Often inside A-MPDU	Primary aggregation method

Typical Implementation: A-MPDU containing 2-64 MPDUs, each potentially an A-MSDU

Numerical Example 2: Aggregation Efficiency Gain I

Problem: Compare efficiency for VoIP packets (160 bytes payload) with and without aggregation.

- 1 Calculate efficiency without aggregation
- 2 Calculate efficiency with A-MPDU aggregating 10 VoIP packets
- 3 What's the throughput improvement?

Given:

- VoIP packet: 160 bytes payload + 40 bytes UDP/IP headers = 200 bytes MSDU
- MAC header: 30 bytes, FCS: 4 bytes = 34 bytes overhead
- PHY preamble: 20 μ s, symbol: 4 μ s
- Data rate: 65 Mbps per spatial stream
- ACK: 14 bytes at basic rate (6 Mbps). Block ACK = 32B
- Aggregation overhead: A-MPDU delimiter (4 bytes) per subframe

Numerical Example 2: Aggregation Efficiency Gain I

Given:

- VoIP: 160B payload + 40B headers = 200B MSDU
- MAC header + FCS = 34B
- A-MPDU delimiter = 4B per MPDU
- PHY preamble = 20 μs
- Data rate = 65 Mbps, Control rate = 6 Mbps
- ACK = 14B, Block ACK = 32B
- DIFS = 34 μs , SIFS = 16 μs , Backoff = 155 μs

1. Without Aggregation

- MPDU size:

$$200 + 34 = 234 \text{ B} = 1872 \text{ bits}$$

Numerical Example 2: Aggregation Efficiency Gain II

- Data Tx time:

$$20 + \frac{1872}{65} = 48.8 \mu s$$

- ACK time:

$$20 + \frac{14 \times 8}{6} = 38.7 \mu s$$

- Total time:

$$34 + 155 + 48.8 + 16 + 38.7 = 292.5 \mu s$$

- Throughput:

$$\frac{160 \times 8}{292.5} = 4.38 \text{ Mbps}$$

2. A-MPDU (10 packets)

- Per MPDU:

$$200 + 34 + 4 = 238 \text{ B}$$

Numerical Example 2: Aggregation Efficiency Gain III

- Total size:

$$10 \times 238 = 2380 \text{ B} = 19040 \text{ bits}$$

- Data Tx time:

$$20 + \frac{19040}{65} = 313.7 \mu s$$

- Block ACK time:

$$20 + \frac{32 \times 8}{6} = 62.7 \mu s$$

- Total time:

$$34 + 155 + 313.7 + 16 + 62.7 = 581.4 \mu s$$

- Throughput:

$$\frac{1600 \times 8}{581.4} = 22.0 \text{ Mbps}$$

3. Improvement

$$\frac{22.0}{4.38} \approx 5.02\times$$

Final Result:

- Without aggregation: **4.38 Mbps**
- With A-MPDU: **22.0 Mbps**
- Gain: $\approx 5\times$ **improvement**

Block Acknowledgments (BlockAck)

- **Problem:** Individual ACKs for aggregated frames inefficient
- **Solution:** Single acknowledgment for multiple frames

BlockAck Process: Block ACK is a mechanism where multiple frames are acknowledged using a single acknowledgment message.

- 1 **Setup:** ADDBA (Add Block Acknowledgment) Request/Response exchange
- 2 **Data Transfer:** Multiple frames sent in aggregation
- 3 **BlockAck Request (BAR):** Sender requests acknowledgment
- 4 **BlockAck (BA):** Receiver responds with bitmap (which frames received)

BlockAck Types:

- **Immediate BlockAck:** BA sent immediately after BAR (low latency)
- **Delayed BlockAck:** BA can be delayed (for compatibility)

Bitmap: 128×16 bit = up to 2048 frames acknowledged in one BA

HT (High Throughput) PHY Enhancements

HT (High Throughput) enhancements are features introduced in 802.11n to increase Wi-Fi speed and efficiency.

- **Guard Interval Options:**

- **Normal GI:** 800 ns (same as 802.11a/g)
- **Short GI:** 400 ns (improves efficiency by 11%)

- **Increased Subcarriers:** Data + Pilot subcarriers

- 20 MHz: 56 subcarriers (52 in 802.11a/g)
- 40 MHz: 114 subcarriers

Used in numericals:
20 MHz → 52
40 MHz → 108

☞ This counts:

- **Higher Modulation:** QAM = Quadrature Amplitude Modulation

Only DATA subcarriers

- 64-QAM standard (same as 802.11a/g)
- Optional: 256-QAM (not widely implemented in 802.11n)

- **MCS (Modulation and Coding Scheme) Tables:**

- 77 MCS indices (0-76)
- Defines: Modulation, coding rate, spatial streams, GI, bandwidth

QAM: It is a modulation technique where amplitude and phase of a signal are varied together to transmit data.

MCS Table Examples (20 MHz)

MCS Index	Modulation	Coding Rate	Spatial Streams	GI=800ns	GI=400ns
0	BPSK	1/2	1	6.5 Mbps	7.2 Mbps
1	QPSK	1/2	1	13.0 Mbps	14.4 Mbps
2	QPSK	3/4	1	19.5 Mbps	21.7 Mbps
3	16-QAM	1/2	1	26.0 Mbps	28.9 Mbps
4	16-QAM	3/4	1	39.0 Mbps	43.3 Mbps
5	64-QAM	2/3	1	52.0 Mbps	57.8 Mbps
6	64-QAM	3/4	1	58.5 Mbps	65.0 Mbps
7	64-QAM	5/6	1	65.0 Mbps	72.2 Mbps
...	...	...	...	...	...
15	64-QAM	5/6	2	130.0 Mbps	144.4 Mbps
23	64-QAM	5/6	3	195.0 Mbps	216.7 Mbps
31	64-QAM	5/6	4	260.0 Mbps	288.9 Mbps

Note: 40 MHz channels approximately double these rates

Beamforming in 802.11n

Beamforming is a technique where signals from multiple antennas are focused toward a specific receiver.

Instead of broadcasting signal everywhere: Focus energy in one direction

- **Concept:** Shape radiation pattern toward receiver
- **How:** Adjust phase and amplitude of each antenna
- **Types:**
 - **Implicit Beamforming:** AP estimates channel from client transmissions
 - **Explicit Beamforming:** Client provides channel feedback to AP
- **Benefits:**
 - Improved SNR at receiver
 - Higher data rates possible
 - Reduced interference to others
- **Limitation:** Often vendor-specific implementations

How it works

- > Adjust phase and amplitude of signals
- > Signals combine constructively at receiver

Greenfield vs Mixed Mode Performance

Mode	Throughput	Compatibility	Use Case
Greenfield	100% (reference)	802.11n only	Enterprise, new deployments
HT Mixed	70-85% of Greenfield	802.11a/g/n	Default, most common
Legacy	20-40% of Greenfield	Full a/b/g/n	Maximum compatibility

Protection Overhead:

- CTS-to-self: Adds $\sim 20\text{-}30\ \mu\text{s}$ per transmission
- Legacy preamble: Longer preambles for legacy compatibility
- Lower basic rates: Control frames sent at slower rates

- **Typical Client Devices:**

- Laptops: 2×2 or 3×3 common
- Smartphones: 1×1 or 2×2
- Tablets: 1×1 or 2×2

- **AP Configurations:**

- Entry-level: 2×2
- Mid-range: 3×3
- Enterprise: 4×4

- **Practical Speeds:**

- 2×2 client to 2×2 AP: 150-200 Mbps TCP
- 3×3 client to 3×3 AP: 200-250 Mbps TCP
- With legacy devices present: 30-50% reduction

Power Save Mechanisms in 802.11n

Power saving in IEEE 802.11n is designed to reduce battery consumption by allowing devices (STAs) to sleep when there's no data and wake up only when needed.

- **Two new dedicated mechanisms** were introduced in 802.11n:
 - Spatial Multiplexing Power Save (SMPS)
 - Power Save Multi-Poll (PSMP)
- **Legacy Power Save enhancement**
 - Still based on the original 802.11 PS mode
 - Optimised for MIMO operation

Legacy Power Save (PS Mode)

How it works:

- Device tells AP → "I am going to sleep"
- AP buffers incoming packets
- Device wakes up periodically
- Checks beacon for pending data
- Sends request → AP delivers data

All mechanisms aim to reduce energy consumption while maintaining high throughput.

Wireless devices don't keep the radio ON all the time. Instead:
Sleep mode → saves power
Wake up periodically → check for data

Spatial Multiplexing Power Save (SMPS)

- Client can disable one or more receive chains
- *Static SMPS*: fixed reduced chain count
- *Dynamic SMPS*: chains turned on/off on demand
- Fewer active chains → lower power consumption

Turn off extra antennas to save power.

Power Save Multi-Poll (PSMP)

- AP schedules wake times for multiple clients
- Clients wake only for their assigned transmission window
- Reduces contention and idle listening
- Well suited for real-time traffic with predictable patterns

AP schedules when each device can transmit/receive.

Enhanced Legacy Power Save

- Uses the same PS-Poll / TIM mechanism as 802.11a/b/g
- Optimised with MIMO awareness:
 - AP buffers data and announces it in the Beacon
 - Client wakes, retrieves frames, then dozes again
- With MIMO, the client can use multiple spatial streams when awake

Key Trade-Off

Power savings vs. *Maximum data rate*

- Fewer active receive chains = less power but lower throughput
- Dynamic SMPS allows balancing the two according to traffic load

802.11n Deployment Considerations

- ❶ **Channel Planning:** 40 MHz only in 5 GHz, avoid in 2.4 GHz
- ❷ **AP Placement:** MIMO benefits from multipath
- ❸ **Client Mix:** Legacy devices significantly reduce performance
- ❹ **Power Settings:** Lower power often better for capacity
- ❺ **MIMO Configuration:** Match AP capabilities to client population

Best Practices:

- Disable 802.11b rates (set minimum to 12 Mbps)
- Use 40 MHz channels only in 5 GHz with sufficient channel separation
- Consider separate SSIDs for legacy vs. 802.11n devices
- Use band steering to move capable clients to 5 GHz

802.11n vs Previous Generations

Metric	802.11g	802.11a	802.11n
Max PHY Rate	54 Mbps	54 Mbps	600 Mbps
Practical TCP	20-25 Mbps	20-25 Mbps	100-300 Mbps
Frequency	2.4 GHz	5 GHz	2.4 & 5 GHz
Key Tech	OFDM	OFDM	MIMO + OFDM
Channel Width	20 MHz	20 MHz	20/40 MHz
Efficiency	Low	Low	High (aggregation)

Performance Improvement: 4-10 $\times$ real-world throughput increase

Limitations of 802.11n

- ① **Client Limitations:** Most clients have fewer antennas than APs
- ② **Interference:** 40 MHz channels more susceptible
- ③ **Legacy Impact:** Mixed mode performance penalty
- ④ **Beamforming:** Vendor proprietary, limited interoperability
- ⑤ **Multi-User:** Still single-user at a time (SU-MIMO)
- ⑥ **Channel Bonding:** Reduces available channels significantly

These limitations led to 802.11ac (next lecture)

Summary: 802.11n Key Innovations

- 1 **MIMO:** Multiple antennas for spatial multiplexing ($2\text{-}4\times$ throughput)
- 2 **Channel Bonding:** 40 MHz channels ($\sim 2\times$ throughput)
- 3 **Frame Aggregation:** A-MSDU and A-MPDU ($2\text{-}4\times$ efficiency)
- 4 **Block Acknowledgments:** Reduced ACK overhead
- 5 **Short Guard Interval:** 400 ns vs 800 ns (11% improvement)
- 6 **Dual-Band:** 2.4 GHz and 5 GHz support
- 7 **Backward Compatibility:** Mixed mode operation

Overall: 802.11n delivered $4\text{-}10\times$ real-world improvement over 802.11a/g

- **Required Reading:**

- Textbook (Gast): Chapter 8 - "802.11n: High Throughput"
- Review MIMO and aggregation concepts

- **Optional Reading:**

- IEEE 802.11n-2009 standard (selected sections)
- MIMO theory fundamentals

- **Next Lecture (Lecture 9): 802.11ac (Wi-Fi 5) - Gigabit Wi-Fi**

- 5 GHz exclusive operation
- Wider channels (80, 160 MHz)
- Multi-User MIMO (MU-MIMO)
- 256-QAM modulation
- Beamforming improvements

Review Questions

- 1 How does spatial multiplexing differ from spatial diversity?
- 2 Why is 40 MHz channel bonding problematic in 2.4 GHz?
- 3 Calculate: 3×3 MIMO with 40 MHz channels at MCS 23, what PHY rate?
- 4 What's the difference between A-MSDU and A-MPDU?
- 5 How does a BlockAck improve efficiency compared to individual ACKs?
- 6 Why does one legacy 802.11g device slow an 802.11n network?
- 7 What's the throughput improvement from Short Guard Interval?
- 8 Why do most clients achieve less than maximum 802.11n speeds?

Discussion Question: "802.11n introduced optional 256-QAM modulation, but it was rarely implemented. Why did vendors skip this optional feature, and why did it become standard in 802.11ac?"

Exercise: MIMO and Aggregation Analysis

- **Objective:** Measure 802.11n performance improvements
- **Tools:** Modern Wi-Fi adapters, Wireshark, iPerf
- **Tasks:**
 - ① Measure throughput for different MIMO configurations (1×1 , 2×2 , 3×3)
 - ② Compare 20 MHz vs 40 MHz channel performance
 - ③ Capture and analyze A-MPDU frames
 - ④ Observe BlockAck exchanges
 - ⑤ Test mixed mode vs greenfield performance
- **Deliverable:** Performance comparison report with analysis

Appendix: MCS Index Calculation

Data Rate Formula:

$$R = \frac{N_{SD} \times N_{BPSCS} \times R_c \times N_{SS}}{T_{SYM}}$$

Where:

- N_{SD} = Number of data subcarriers (52 for 20 MHz, 108 for 40 MHz)
- N_{BPSCS} = Bits per subcarrier (1 for BPSK, 2 for QPSK, 4 for 16-QAM, 6 for 64-QAM)
- R_c = Coding rate (1/2, 2/3, 3/4, 5/6)
- N_{SS} = Number of spatial streams
- T_{SYM} = Symbol duration (4 μ s for normal GI, 3.6 μ s for short GI)

Example (MCS 7, 20 MHz, 1 stream, normal GI):

$$R = \frac{52 \times 6 \times 5/6 \times 1}{4 \times 10^{-6}} = \frac{52 \times 5}{4 \times 10^{-6}} = 65 \text{ Mbps}$$

Appendix: Common 802.11n Configurations

Device Type	Typical MIMO	Max PHY Rate	Practical TCP	Notes
Smartphone	1×1 or 2×2	65-150 Mbps	45-105 Mbps	Size/power constrained
Tablet	1×1 or 2×2	65-150 Mbps	45-105 Mbps	Similar to phones
Laptop	2×2 or 3×3	150-300 Mbps	105-210 Mbps	Most common
Desktop Adapter	2×2 or 3×3	150-300 Mbps	105-210 Mbps	External antennas
Entry AP	2×2	300 Mbps	210 Mbps	Home/SMB
Mid AP	3×3	450 Mbps	315 Mbps	Prosumer
Enterprise AP	4×4	600 Mbps	420 Mbps	Business

Note: Practical TCP is ~70% of PHY rate assumption

Optimal Aggregation Size Depends On:

① Traffic Type:

- Bulk transfer: Large aggregates (64 KB)
- Interactive: Small aggregates (4-8 KB)
- VoIP: Very small or no aggregation

② Channel Conditions:

- Good SNR: Large aggregates
- Poor SNR: Small aggregates (less lost on error)

③ Buffer Availability: Both transmitter and receiver

Default Values:

- A-MPDU maximum: 64 KB (65,535 bytes)
- A-MSDU maximum: 8 KB (7,935 bytes)
- Typical aggregates: 8-32 KB

Appendix: 802.11n Timeline

- **2002:** Work begins on High Throughput Task Group (TGn)
- **2004:** Competing proposals (WWiSE vs TGn Sync)
- **2006:** Draft 2.0 products hit market (pre-standard)
- **2007:** Apple includes 802.11n in iPhone (draft)
- **2009:** Final standard ratified (7 years after starting!)
- **2010:** Wi-Fi Alliance begins certification
- **2012:** Most new devices 802.11n capable
- **2014:** Still dominant technology (before 802.11ac took over)

Lesson: Long standardization process (7 years), but draft products deployed early.

Appendix: Real-World Performance Measurements

Test Setup: 2×2 MIMO, 40 MHz, 5 GHz, no interference

- **FTP Transfer (large files):** 120-180 Mbps TCP
- **Web Browsing (mixed):** 60-100 Mbps effective
- **VoIP (no aggregation):** 5-10 Mbps per call
- **Video Streaming:** 40-80 Mbps (multiple streams)

Factors Reducing Performance:

- 1 Legacy devices on network (30-50% reduction)
- 2 Interference (20-40% reduction)
- 3 Distance (exponential reduction)
- 4 Many clients sharing (fair sharing reduces per-client)

Thank you!